Neuro-Mathematical Transfer Engine (NMTE): A Mathematical Framework for Cross-Embodiment Sim-to-Real Adaptation in Industrial Robotics

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***Abstract*—The deployment of robotic arms and quadrupedal robots in dynamic industrial environments—such as high-throughput warehouses and unpredictable railway tracks—hinges on the efficacy of sim-to-real transfer learning. Despite advancements in Deep Reinforcement Learning (DRL) and domain randomization, transferring policies across disparate physical realities and kinematic embodiments remains a critical bottleneck. This paper extends the analysis of contemporary research gaps to propose the mathematical architecture of a " Neuro-Mathematical Transfer Engine (NMTE)." By integrating Koopman Operator Theory for invariant dynamics modeling, Gromov-Wasserstein Optimal Transport for cross-embodiment latent alignment, and Disturbance Observer-based Control Barrier Functions (CBFs) for certifiable safety, we formulate a comprehensive research problem. The proposed transfer engine provides a theoretically grounded framework to bridge the persistent reality gap while adhering to edge-compute constraints.**

Keywords—Transfer Learning, Sim-to-Real, Optimal Transport, Koopman Operator, Control Barrier Functions, Cross-Embodiment.

I. INTRODUCTION

The integration of intelligent robotic systems in unstructured domains requires rapid, generalized decision-making. While training Deep Reinforcement Learning (DRL) policies in physics simulators effectively circumvents the cost of real-world data collection, transferring these policies to physical hardware introduces a severe degradation in performance, known as the "reality gap" [7], [11]. In warehouse logistics and railway inspection, robots face dynamic loads, unpredictable friction, and highly variable terrain constraints that invalidate standard simulated approximations [8], [15].

Traditional sim-to-real techniques largely rely on Domain Randomization (DR) and task-specific fine-tuning [7], [9]. However, as noted by Jaquier et al. [11], empirical heuristics scale poorly across heterogeneous robot fleets. The necessity to manually retune policies for every new embodiment (e.g., from a 6-DoF robotic arm to a quadrupedal manipulator) represents a major barrier to multi-robot orchestration [1], [12]. To resolve these limitations, this paper first synthesizes prevailing research gaps from state-of-the-art literature (2023–2026). Subsequently, it proposes the development of a " Neuro-Mathematical Transfer Engine (NMTE)," a unified mathematical framework utilizing functional analysis and optimal transport to automate cross-embodiment sim-to-real adaptation safely and efficiently.

II. RESEARCH GAPS IN SIM-TO-REAL TRANSFER

Our review of recent literature identifies five interconnected gaps preventing scalable industrial deployment:

A. Gap 1: Zero-Shot Sim-to-Real Transfer in High-Precision, Contact-Rich Manipulation

In modern warehouses, robotic arms are frequently tasked with complex object stacking, bin picking, and shelf placement. Transferring these contact-rich manipulation skills from simulation to reality without extensive real-world fine-tuning remains elusive.

He et al. [1] recently demonstrated that applying Transfer Learning (TL) to evolutionary algorithms (specifically iCEM+TL) can improve success rates in low-level motion planning by up to 23%. However, the same study [1] reveals that their zero-shot transfer framework struggles significantly in complex geometric scenarios requiring sliding and tight shelf placement, highlighting a persistent gap in modeling hard-contact dynamics.

Similarly, Saad et al. [2] proposed integrating Large Language Models (LLMs) with RL to bridge high-level reasoning and low-level control. Although their framework achieved a 33.5% decrease in task completion time for a Franka Emika Panda arm [2], the authors explicitly note in [2] that direct sim-to-real transfer of the RL-driven low-level control remains a severe bottleneck due to unpredictable friction.

Furthermore, Ding [3] highlights that applying pure DRL to a 6-DoF robotic arm suffers from convergence instability and sparse rewards in high-dimensional spaces. While [3] utilizes stepwise reward shaping combined with Inverse Kinematics (IK) to mitigate this, the methodology still requires deterministic environmental assumptions. To enhance physical realism, Elseiagy et al. [8] showed that automated dataset generation utilizing Transformers can improve dynamic parameter estimation. Yet, as acknowledged in [8], computing these dynamic parameters in real-time during contact-rich tasks

imposes heavy latency. Thus, a fundamental gap remains in achieving reliable zero-shot transfer for high-precision, contact-rich manipulation without latency-inducing dynamic parameter recalculations [1], [3], [8].

B. Gap 2: Dynamic Task-Invariant Feature Extraction for Locomotion on Degraded Terrains

Quadruped robots deployed for railway inspection and construction site monitoring face continuously shifting terrains (e.g., loose ballast, mud, rubble). Most domain adaptation techniques rely heavily on static domain randomization.

Shakerimov et al. [6] demonstrate that while domain randomization helps, the direct transition to the real world causes a "distributional shift" problem, leading to erroneous Q-value estimations. To combat this, [6] suggests hybrid fine-tuning on the target domain; however, this negates the benefits of pure simulation training.

Attempting to circumvent this, Liang et al. [4] introduce the DreamTIP framework, which extracts "Task-Invariant Properties" (such as contact stability and terrain clearance) using a Dreamer world model. Although DreamTIP successfully masks dynamic variations [4], the reliance on Large Language Models to identify these properties [4] introduces a high dependency on visual pre-processing, which fails under visually degraded conditions.

Exploring non-visual modalities, de Kam et al. [5] utilize tactile suction-cup feet on a soft quadruped, demonstrating a 91% forward speed increase on inclines using closed-loop tactile feedback. Yet, the authors in [5] note that transferring tactile force estimates from simulation is exceptionally noisy. Su et al. [13] approach the problem mathematically, demonstrating that leveraging morphological symmetries in RL can dramatically stabilize legged locomotion. Nevertheless, unstructured terrains inherently break these symmetrical assumptions [13]. Consequently, there is a critical gap in developing multi-modal (tactile-proprioceptive) task-invariant feature extractors that do not rely on perfect symmetrical priors [13] or noise-free simulation data [5], [6].

C. Gap 3: Parameter-Efficient Cross-Embodiment Representation Learning

A modern logistics hub utilizes a heterogeneous fleet of robots, ranging from fixed manipulators to quadrupedal inspectors. Currently, policies are trained in silos.

Jaquier et al. [7] point out that the robotics community urgently needs "cross-embodiment" datasets to train models that can master various embodiments. Unfortunately, [7] notes that aligning features between drastically different kinematic chains (e.g., an arm vs. a quadruped) is an unresolved mathematical challenge.

Addressing system-level architectures, the authors of NavRL++ [15] achieved zero-shot sim-to-real transfer across both aerial and legged robots for navigation tasks. However, as noted in [15], while navigation features transfer reasonably well, this system-level framework fails to transfer fine-grained manipulation or interaction features.

Zhang [11] illustrates the use of transfer learning with an enhanced YOLOv8 algorithm to improve keypoint detection on lightweight robotic arms, achieving a 3.8% mAP increase. While [11] successfully extracts generalized features from a human pose dataset to a robotic arm, applying this visual-spatial intuition to a quadruped's perception module has not been explored. Saad et al. [2] mention that multi-robot scalability is the next frontier for LLM-enhanced RL; however, a parameter-efficient mathematical bridge that aligns the latent spaces of a quadruped's locomotion policy [15] with an arm's manipulation policy [11] is currently missing from the literature [2], [7].

D. Gap 4: Certifiable Safety Guarantees in Continuous Sim-to-Real Domain Adaptation

Safety is paramount when deploying robots in environments shared with humans, such as active railways and collaborative warehouses. Transfer learning often acts as a "black box," making it difficult to guarantee safety post-adaptation.

A 2025 comprehensive review in Robotics [9] explicitly identifies that ensuring neural policies can recover from perturbations without compromising safety is a "pressing research direction," especially because current learning-based locomotion lacks verifiable safety guarantees. The review [9] stresses that reachability analysis and SMT solvers scale poorly to high-dimensional quadruped policies.

Cruz Ulloa et al. [10] deployed quadruped robots in unpredictable Search and Rescue (SAR) and post-disaster scenarios, focusing on maximizing efficiency. Yet, [10] highlights that finding optimal operational solutions often comes at the cost of bounding the robot's unpredictable exploratory actions.

To inject stability, Bellegarda and Ijspeert [14] developed CPG-RL, successfully merging biologically inspired Central Pattern Generators (CPGs) with RL. While CPG-RL provides excellent inductive biases for rhythmic stability [14], it does not inherently prevent unsafe joint torques during sudden sim-to-real domain shifts. Ding [3] attempted to solve safety in arms through stepwise reward shaping, implicitly teaching the agent safe boundaries. However, reward shaping [3] does not offer mathematical safety certification. Therefore, establishing "certifiable" Source-Free Domain Adaptation (SFDA) that mathematically guarantees constraint satisfaction during real-time domain shifts remains an open problem [9], [10], [14].

E. Gap 5: Energy-Aware and Compute-Constrained Transfer Learning on Edge Devices

Industrial robots, particularly autonomous quadrupeds and mobile manipulators, operate under strict battery and edge-compute constraints. Modern transfer learning methods are heavily over-parameterized.

A 2026 review by Patrashko and Gurau [12] reveals that only 23% of machine learning-powered robotic inspection systems achieve full industrial deployment. The study [12] attributes this largely to the massive labeled data and computational power required to process complex unstructured patterns on the edge.

Zhang [11] sought to resolve edge limitations by drastically reducing floating-point operations in YOLOv8, proving that transfer learning can make models more suitable for

environments with limited computational capacity. Despite this success in vision [11], applying similar computational pruning to continuous control policies remains difficult.

Furthermore, while He et al. [1] focus on "sample-efficient" planning, sample efficiency in simulation does not equate to "energy-efficient" execution on physical hardware [1]. For instance, de Kam et al. [5] utilized pneumatic actuators on a soft quadruped, but mapping neural commands to hardware without optimizing for pneumatic energy loss resulted in rapid battery drain [5]. Similarly, Elseiagy et al. [8] relied on computationally heavy Transformer models for dynamic parameter learning, which [8] admits is difficult to run continuously on a mobile robot's onboard CPU. The literature critically lacks frameworks that jointly optimize for sim-to-real task success and real-time physical energy consumption within edge-compute constraints [5], [11], [12].

III. Architecture of the Neuro-Mathematical Transfer Engine (NMTE)

A critical limitation of standard Deep Reinforcement Learning (DRL) in sim-to-real transfer is its "black-box" nature, which relies purely on data-driven approximations and lacks out-of-distribution guarantees. Conversely, traditional mathematical control theory fails in unstructured environments because it cannot process high-dimensional sensory data (e.g., LiDAR, vision).

To address the aforementioned gaps, the proposed Neuro-Mathematical Transfer Engine (NMTE) bridges this divide by utilizing "Math-Guided Neural Networks." In this architecture, Deep Neural Networks are strictly utilized as high-dimensional feature extractors and state-encoders, processing unstructured warehouse perception. However, rather than allowing the neural network to predict outputs blindly, the network's latent representations and final actions are constrained by rigorous mathematical priors (Optimal Transport, Koopman Dynamics, and Control Barrier Functions). This hybrid paradigm ensures that the flexibility and scalability of Deep Learning are preserved, while the actual physical execution remains mathematically certifiable and physically bounded. This engine decouples task intention from morphological and physical constraints, dynamically bridging the sim-to-real domain shift through rigorous mathematical priors [1], [18], [24].

A. Koopman Operator Theory for Task-Invariant Dynamics (Gaps 1 & 2)

The reality gap is fundamentally a mismatch in non-linear dynamics $x_{t+1} = F(x_t)$ between simulation and reality [24]. In warehouse manipulation [22] and quadruped locomotion [28], estimating the exact non-linear discrepancy is computationally intractable on edge devices.

The Transfer Engine utilizes Koopman Operator Theory [18], [27] to linearize these dynamics. Let the true real-world dynamics be defined on a state space M. We introduce a learned observable lifting function $g: M \to H$ that maps the finite-dimensional state into an infinite-dimensional Hilbert space H. The Koopman operator K acts linearly on these observables:

$$g(x_{t+1}) = Kg(x_t)$$

In our Transfer Engine, K is learned entirely in simulation to represent the pure "task dynamics" (e.g., maintaining a stable gait or tracking a manipulation trajectory) [18]. During real-world deployment, K remains frozen. We only fine-tune the comparatively lightweight observation encoder $g_\theta(x)$ using real-world sensory inputs [27], [28]. By restricting the sim-to-real fine-tuning to the lifting function g_θ, the Transfer Engine absorbs real-world contact variations (e.g., sliding friction, loose ballast) without requiring recalculation of the core policy [18], [27].

B. Optimal Transport for Cross-Embodiment Latent Alignment (Gap 3)

To enable zero-shot skill transfer between heterogeneous robots (e.g., a manipulator arm identifying tasks for a quadruped [1], [25]), the Transfer Engine maps incompatible state-action spaces into a shared morphological-agnostic manifold.

We model the source embodiment (simulation/arm) and target embodiment (reality/quadruped) as metric measure spaces (X, d_X, μ_S) and (Y, d_Y, μ_T) respectively. Standard domain adaptation fails because the state vectors possess different dimensions [15]. Thus, the engine utilizes the Gromov-Wasserstein (GW) Optimal Transport distance, which aligns the topological structure of the metric spaces rather than absolute coordinates [1], [15]. The GW distance minimizes the cost over all possible couplings $\pi \in \Pi(\mu_S, \mu_T)$:

$$\mathcal{W}_{GW}(\mu_S, \mu_T) = \inf_{\pi} \int_{X\times Y} \int_{X\times Y} \left| d_X(x, x') - d_Y(y, y') \right|^2 d\pi(x, y)\, d\pi(x', y')$$

By minimizing W_{GW} during pre-training, the Transfer Engine establishes a bijective mapping $\Gamma: X \to Y$ between the latent skill primitives [15], [23]. A policy synthesized for an arm extracting a package can thus be mathematically projected onto a quadruped utilizing a mobile manipulator, sharing universal manipulation behaviors without redundant re-training [1], [25].

C. Disturbance-Observer Control Barrier Functions for Safety (Gap 4)

Safety certification breaks down during sim-to-real transfer due to unmodeled dynamics [16], [17]. If a quadruped operating in a railway slips, its standard DRL policy cannot formally guarantee non-collision.

The Transfer Engine enforces safety using a Disturbance Observer-based Control Barrier Function (DOB-CBF) [16]. Let the safe operating region be defined as a superlevel set $C = \{x \in \mathbb{R}^n : h(x) \geq 0\}$. To guarantee forward invariance of C in the physical world, the system requires $\dot{h}(x, u) \geq -\alpha(h(x))$, where α is a class-K function [17].

Because the simulation dynamics $f_{sim}(x)$ differ from real dynamics $f_{real}(x)$, we model the sim-to-real gap as an unknown time-varying disturbance $d(t)$ [16]. The Transfer Engine estimates this bounded discrepancy online via a residual disturbance observer $\hat{d}(t)$. The DRL action u_{RL} is passed through a minimally invasive Quadratic Program (QP) safety filter [16], [17]:

$$u_{safe} = argmin_u \|u - u_{RL}\|^2$$

$$s.t. \nabla h(x) \cdot \left(f_{sim}(x) + g(x)u + \hat{\mathrm{d}}(\mathrm{t})\right) \geq -\alpha\left(h(x)\right)$$

This provides a strict mathematical certificate that constraints (e.g., torque limits [19] or spatial bounds [17]) are satisfied in the real world, actively overriding the RL policy only when $d(t)$pushes the robot toward a critical failure boundary [16], [17].

D. Energy-Aware Edge Execution (Gap 5)

Finally, industrial transfer learning must operate under the strict microsecond constraints of real-time control [19], [26]. While the offline Optimal Transport alignment [15] and Koopman K matrices [18] are computationally heavy during training, their inference counterparts are linear and highly efficient [26], [27]. Furthermore, by projecting high-dimensional perception via the frozen GW mapping, the onboard CPU is only required to compute the lightweight DOB-CBF quadratic program [16] and the Koopman lifting network g_θ [18]. This specific architectural pipeline implicitly regularizes computational load, translating sample efficiency into physical energy efficiency suitable for untethered deployments [19], [26].

IV. Research Roadmap

Building upon the mathematical architecture, development of a "Universal, Safety-Certifiable Neuro-Mathematical Transfer Engine (NMTE) for Cross-Embodiment Sim-to-Real Adaptation in Industrial Robotics via Optimal Transport and Koopman Dynamics" is proposed.

A. Divergence-Free Dynamic Modeling:

Design and validate the Koopman lifting function g_θ to extract task-invariant flow fields for contact-rich warehouse manipulation [22] and unstructured railway locomotion [28], eliminating the need for heavy, edge-taxing domain randomization [18], [27].

B. Latent Manifold Alignment:

Formulate a sample-efficient Gromov-Wasserstein optimization algorithm capable of zero-shot cross-embodiment transfer [1] of low-level manipulation skills between simulated 6-DoF arms and real-world mobile quadrupeds [15], [25].

C. Sim-to-Real Safety Certification:

Integrate a real-time Disturbance Observer (DOB) with Minkowski-based Control Barrier Functions [16], [17] to guarantee collision avoidance and torque-limit safety during the initial, highly uncertain phases of physical deployment [19].

D. Hardware Validation and Edge Optimization:

Prove the computational viability of the Transfer Engine by deploying it on physical edge-compute modules [26], mapping policies trained on high-performance clusters to physical hardware operating at >500 Hz control loops [19], [23].

V. Conclusion

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